

Per-Model Aggregator Comparison:
Centered Clipping, Geometric Median, Multi-Krum, Trimmed Mean

ByzFL SNN Benchmark

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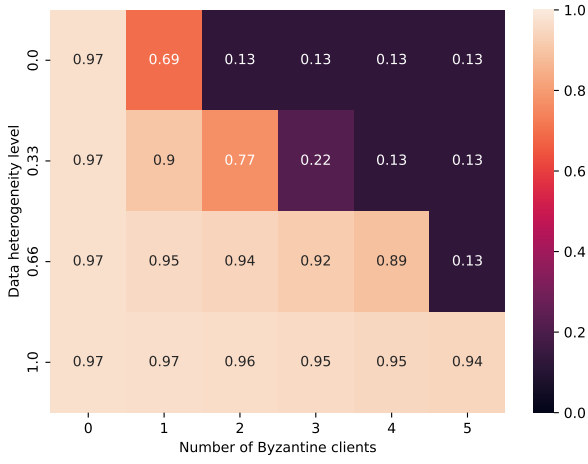
1 Introduction

This report provides a grid of heatmaps for each model architecture, showing the worst-case test accuracy across all attacks for four different robust aggregators:

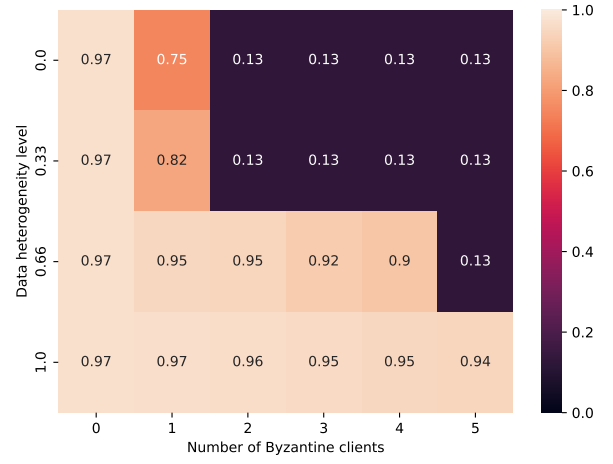
1. Centered Clipping
2. Geometric Median
3. Multi-Krum
4. Trimmed Mean

This allows for a direct visual comparison of which aggregators perform best for Sparse vs. Dense gradients.

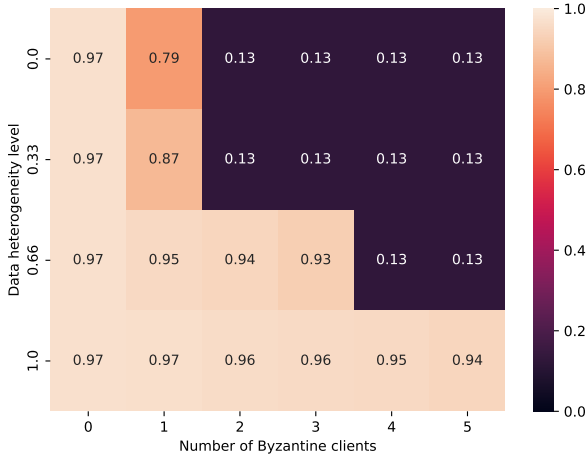
2 SNN Atan ($\alpha = 1.2$)



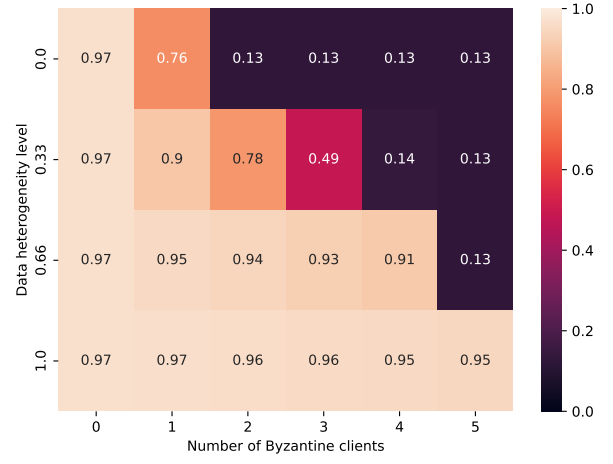
(a) Centered Clipping



(b) Geometric Median



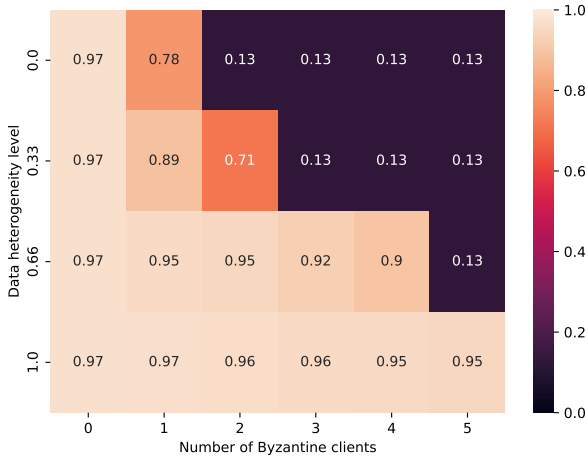
(c) Multi-Krum



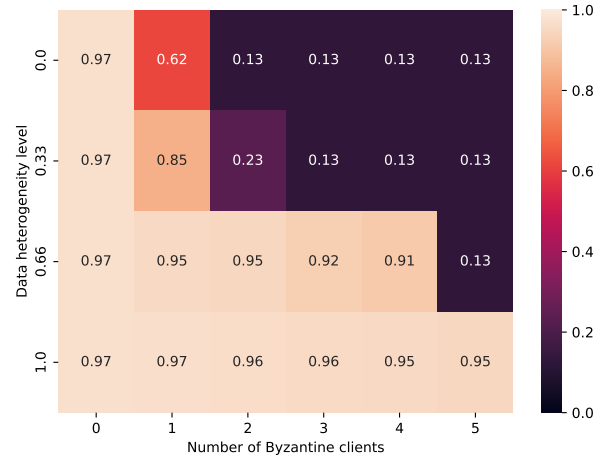
(d) Trimmed Mean

Figure 1: SNN Atan ($\alpha = 1.2$) - Worst-case accuracy across all attacks for different aggregators.

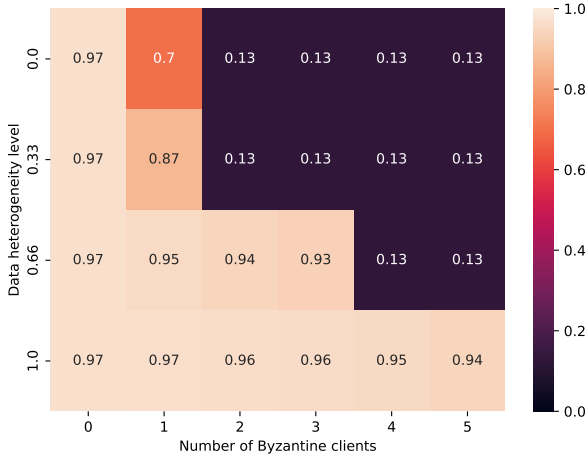
3 SNN Tri ($\beta = 2.0$)



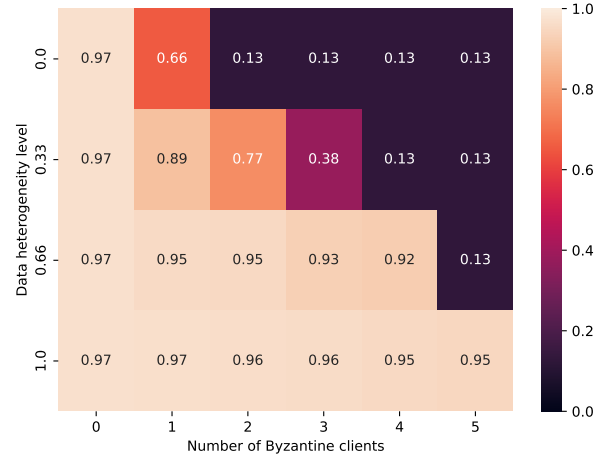
(a) Centered Clipping



(b) Geometric Median



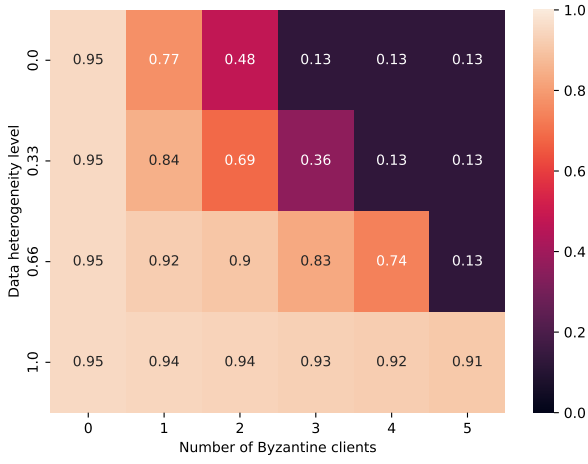
(c) Multi-Krum



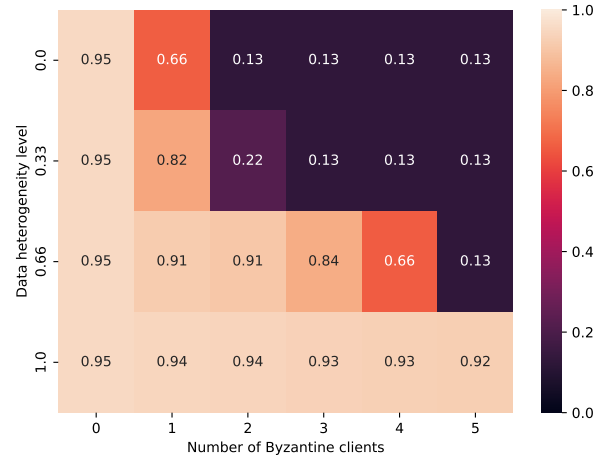
(d) Trimmed Mean

Figure 2: SNN Tri ($\beta = 2.0$) - Worst-case accuracy across all attacks for different aggregators.

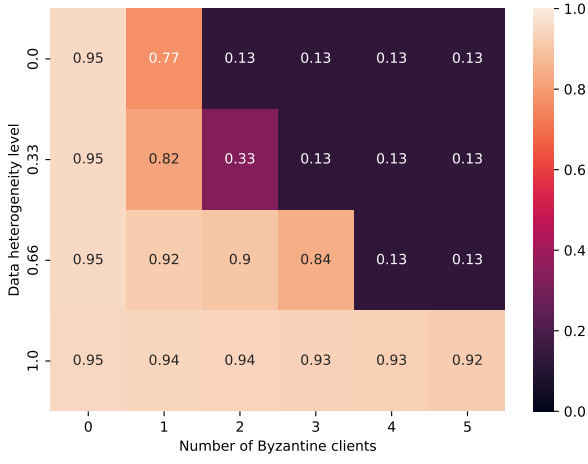
4 SNN Box ($\beta = 2.0$)



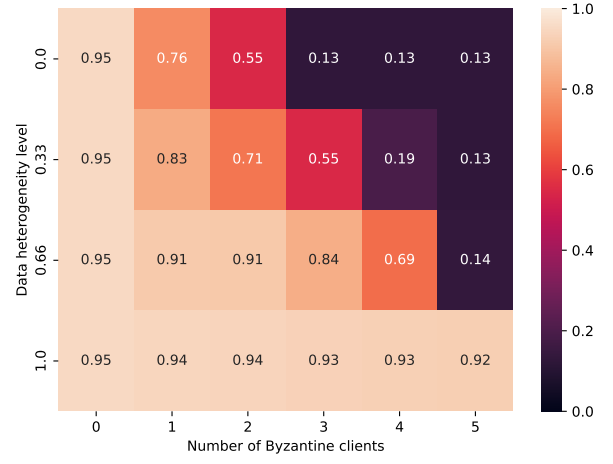
(a) Centered Clipping



(b) Geometric Median



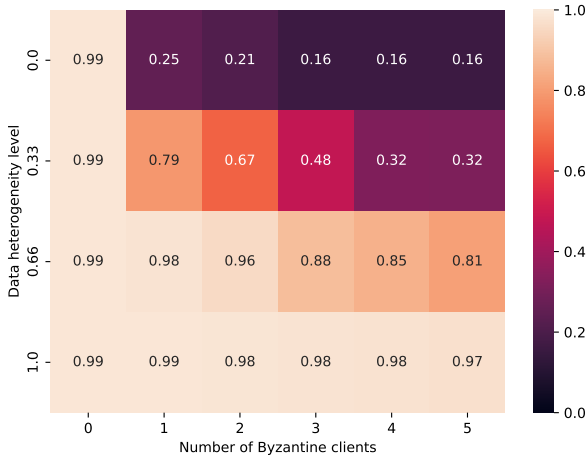
(c) Multi-Krum



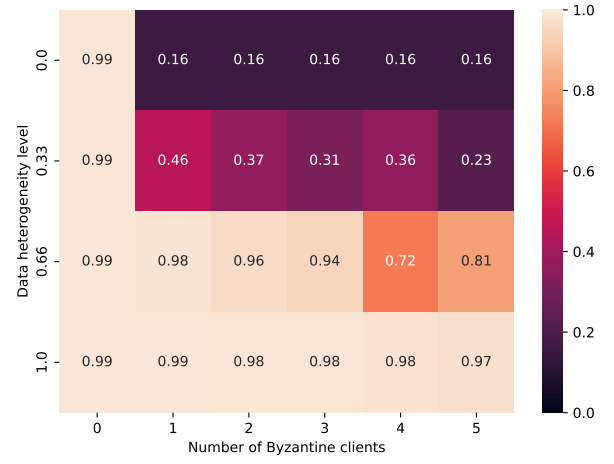
(d) Trimmed Mean

Figure 3: SNN Box ($\beta = 2.0$) - Worst-case accuracy across all attacks for different aggregators.

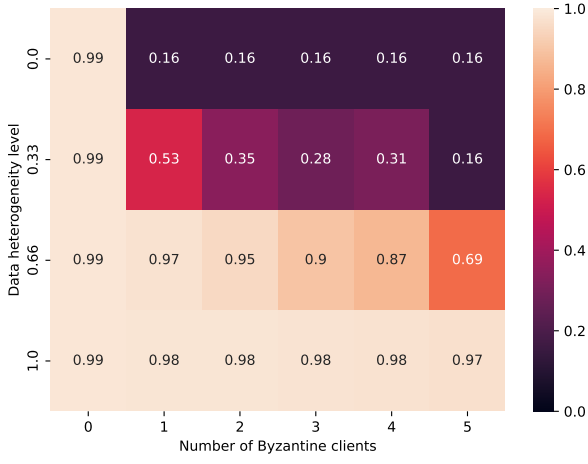
5 CNN ReLU (Baseline, $LR = 0.15$)



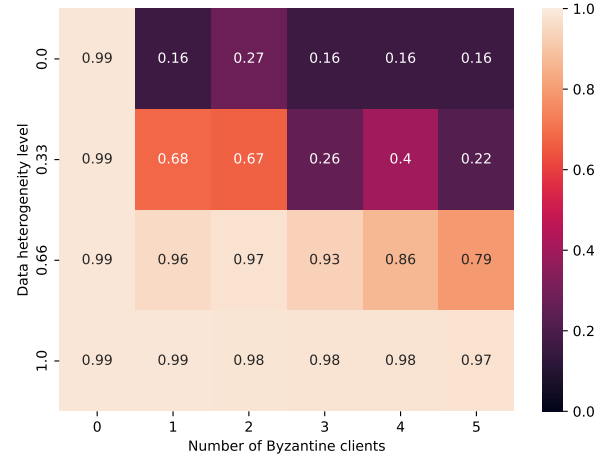
(a) Centered Clipping



(b) Geometric Median



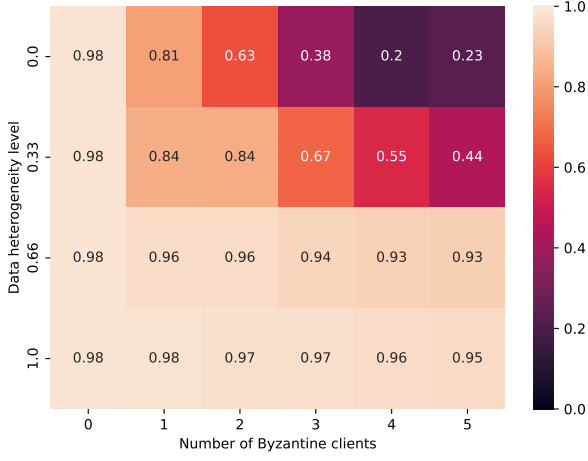
(c) Multi-Krum



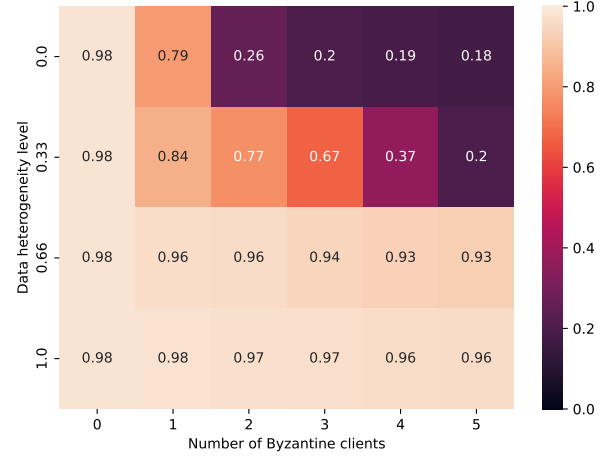
(d) Trimmed Mean

Figure 4: CNN ReLU (No Clip, $LR = 0.15$) - Worst-case accuracy across all attacks for different aggregators.

6 CNN Tanh



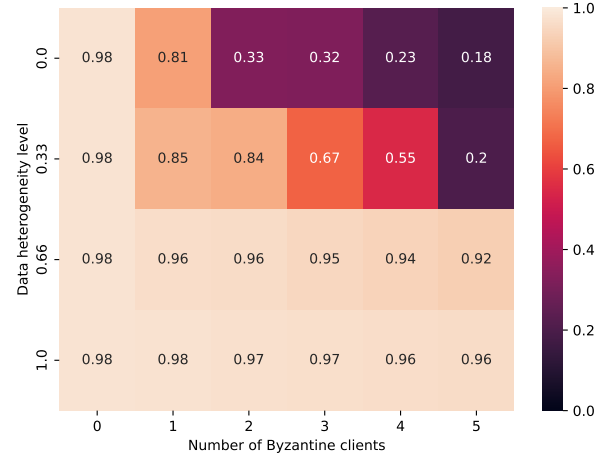
(a) Centered Clipping



(b) Geometric Median



(c) Multi-Krum



(d) Trimmed Mean

Figure 5: CNN Tanh - Worst-case accuracy across all attacks for different aggregators.

7 CNN ReLU (Gradient Clipped)



Figure 6: Geometric Median (Only available aggregator)
CNN ReLU (Grad Clipped) - Worst-case accuracy.